

CIS241

System-Level Programming and Utilities

bash Conditionals

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Based on material provided by Erin Carrier, Austin Ferguson, and Katherine Bowers

if

General form:

```
if CONDITION
then
    # do the things
fi
```

or:

```
if CONDITION; then
    # do the things
fi
```

CONDITION?

Think of you `if` tests - those are conditionals

```
if [ $1 -lt 10 ]; then  
    # do things here  
fi
```

What does this do?

MORE

Operator	Description
! EXPRESSION	The EXPRESSION is false.
-n STRING	The length of STRING is greater than zero.
-z STRING	The length of STRING is zero (ie it is empty).
STRING1 = STRING2	STRING1 is equal to STRING2
STRING1 != STRING2	STRING1 is not equal to STRING2
INTEGER1 -eq INTEGER2	INTEGER1 is numerically equal to INTEGER2
INTEGER1 -gt INTEGER2	INTEGER1 is numerically greater than INTEGER2
INTEGER1 -lt INTEGER2	INTEGER1 is numerically less than INTEGER2
-d FILE	FILE exists and is a directory.
-e FILE	FILE exists.
-r FILE	FILE exists and the read permission is granted.
-s FILE	FILE exists and its size is greater than zero (ie. it is not empty).
-w FILE	FILE exists and the write permission is granted.
-x FILE	FILE exists and the execute permission is granted.

Conditions

- Can also run `man test`

Alternative - arithmetic expression!

```
if ( ( $1 < 10 ) ); then  
    # do things  
fi
```

And then, the else

```
if CONDITION; then
    # do thing 1
elif CONDITION2; then
    # do thing 2
else
    # do thing 3
fi
```

Boolean operators

```
if [ $1 -lt 10 ] && [ $1 -gt 0 ]; then
    # do stuff
fi
```

```
if [ -z $2 ] || [ $2 = "start" ]; then
    # do stuff
fi
```

And finally, the case

```
test1="HELLO"

case $test1 in
HELLO | HI)
    echo "case successful: $test1" ;;
ANOTHER)
    echo "a different case" ;;
"AND ANOTHER")
    echo "different yet again" ;;
*)
    echo "default case" ;;
esac
```

<https://linuxize.com/post/bash-case-statement/>